Beyond Visibility: Persistent Graph Memory for Visual Social Navigation

Anonymous CVPR submission

Abstract

Pedestrians can remain relevant to navigation after leaving a robot's camera view. A visual policy therefore needs to use recent observations without treating missing or outdated information as current evidence. We present ***Social-STG****, a visual social-navigation framework with persistent spatio-temporal graph memory. A memory bank maintains pedestrian records and temporarily retains unmatched observations as latent human nodes. Edge-conditioned attention models their interactions with the robot and goal, while uncertainty-gated updates regulate how messages enter recurrent states associated with each record. The pooled graph representation is fused with the visual policy, and auxiliary tasks provide training signals for motion, visibility, and interaction cues. On Social-HM3D and zero-shot Social-MP3D, SocialSTG improves task completion and success-weighted efficiency over Falcon and classical planners while reducing human-collision episodes. Ablations support complementary roles for record retention, temporal gating, and graph input to the policy; motion features and graph-based auxiliary supervision mainly benefit completion and efficiency. Scene-specific adaptation in a reconstructed campus cafeteria yields further gains, with SocialSTG retaining an advantage over adapted Falcon. These gains do not extend uniformly to personal-space compliance. The results support persistent interaction memory as a useful addition to visual social navigation, while leaving the balance between goal progress and interpersonal clearance an open challenge.*

1. Introduction

A pedestrian who leaves a robot's camera view can still affect its next action. In a corridor or cafeteria, the robot may turn away from someone, furniture may interrupt the line of sight, or another person may briefly obscure a detection. These changes in visibility do not end the interaction [29, 37]. A visual navigation policy must decide how to use current observations while preserving relevant information from the recent past. This makes short-term pedestrian memory an important part of socially aware navigation [4, 11, 13, 24].

Future-aware policies such as Falcon [13] learn from human-motion prediction, while relational crowd models [4, 24] and occlusion-aware planners [31, 45, 47] offer ways to reason about interactions and incomplete observations. These approaches motivate a related question for visual policies: *what information should remain available about a recently unobserved pedestrian, and how should its influence change as the observation becomes older?* An explicit record can make this history accessible to the policy, but retaining it indefinitely or trusting it as much as a fresh observation may introduce misleading information.

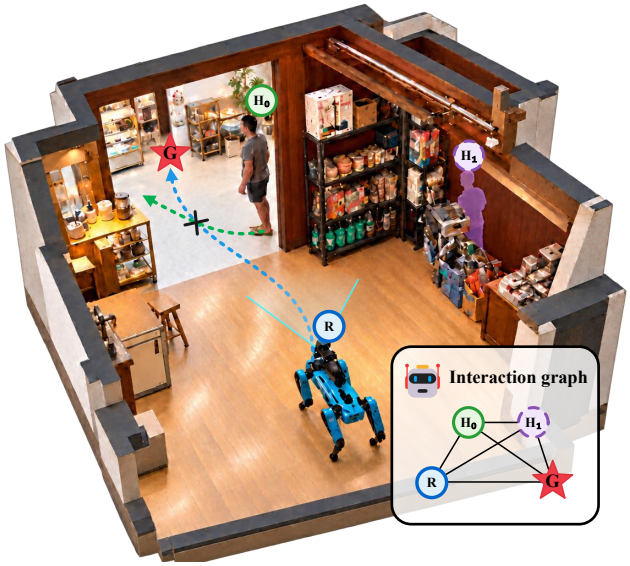


Figure 1. **Social navigation with short-term interaction memory.** The cutaway scene is a conceptual illustration informed by Habitat observations. A robot (R) approaches a goal (G) amidst visible (H_0) and occluded (H_1) pedestrians outside its camera FoV (cyan). Overlaid curves and $\times$ denote projected paths and a potential crossing. The inset graph models these relations, using a dashed node to represent the retained memory of the unobserved pedestrian.

The challenge is therefore both to preserve useful history and to control how it affects new decisions. Pedestrian detections appear and disappear as the robot moves, changing the set of active graph nodes. A recurrent representation must remain associated with the same memory record as node slots change. Meanwhile, older observations become less reliable, so persistence needs to be paired with a mechanism that can attenuate uncertain messages rather than simply

accumulating them.

We introduce **SocialSTG**, which adds persistent spatio-temporal graph memory to a Falcon-based visual policy [13], using the same pretrained PointNav initialization. A position-based memory bank associates detections with stored records and retains unmatched records for a bounded lifetime. The graph represents current and retained human records together with the robot and goal. Edge-conditioned spatial attention encodes their relations, and uncertainty-gated temporal updates integrate messages into record-indexed recurrent states. A robot-centric pooling layer summarizes the graph for residual fusion into the policy. During training, auxiliary tasks supervise motion, visibility, and interaction cues in the human-node representations.

Experiments on Social-HM3D and zero-shot Social-MP3D show higher completion and success-weighted efficiency, together with fewer human-collision episodes, than Falcon, A*, and ORCA. Personal-space compliance follows a different pattern: ORCA maintains greater clearance, and SocialSTG does not improve every social metric. The ablations help clarify the source of the navigation gains. Removing record retention produces the largest degradation in completion and the highest collision rate among the variants, while removing temporal gating or graph input also worsens both outcomes. Motion features and graph-based auxiliary supervision contribute primarily to completion and efficiency. In a reconstructed campus cafeteria, scene-specific adaptation improves both Falcon and SocialSTG, with SocialSTG retaining an advantage in completion, efficiency, and collisions. Together, these results support the use of explicit interaction history without equating improved task performance with uniformly better social behavior.

Our contributions are:

- **Persistent pedestrian memory.** We introduce a bounded memory bank that retains recently unobserved pedestrians as graph records and preserves their recurrent states as the active node set changes.
- **Graph reasoning for visual navigation.** We combine edge-conditioned spatial attention, uncertainty-gated temporal updates, and residual graph-policy fusion, with auxiliary supervision that shapes human-node representations during training.
- **Evaluation of memory and interaction reasoning.** We study benchmark navigation, zero-shot transfer, and cafeteria adaptation, and use component ablations to examine how retention, gating, graph input, and supervision affect completion, efficiency, and social outcomes.

2. Related Work

PointGoal Navigation and Modern Visual Control. Visual PointGoal navigation—reaching target coordinates from egocentric streams—initially relied on large-scale RL (DD-PPO [43], VER [44]) and visual pretraining (OVRL-V2 [46], VC-1 [27]). Recent work bifurcates into generative trajectory modeling and metric-grounded predictive control. Generative policies—spanning foundation models (GNM [35], ViNT [36]) and diffusion, flow-matching, or world-model frameworks (Diffusion Policy [8], NoMaD [40], NaviDiffusor [50], FlowNav [12], NWM [1], NavDP [2])—use action-chunking to capture multimodal path distributions. Concurrently, regression and world-action architectures—from neural mapping [3] and working-memory control (MemoNav [22]) to LoGoPlanner [32] and WAM-Nav [48]—couple observation history with metric geometry or latent foresight. However, these paradigms mainly target static scenes or short-horizon avoidance, lacking explicit relational structures to track occluded dynamic agents over time.

Socially-Aware Navigation and Foundation Models. Socially compliant navigation evolved from RL/IL collision-avoidance proxies [4, 7, 41] and demonstration benchmarks [19] to crowd-group modeling [10, 20], relational graphs (RGL [5], DSRNN [24], HEIGHT [25]), and VQ-VAE trajectory optimization [21]. Recently, Vision-Language Models (VLMs) have been adopted to inject social commonsense priors (VLM-Social-Nav [38], GSON [26], SocialNav [6]). Yet, despite progress in occlusion inference [31] and stochastic occupancy mapping [45], classical and VLM policies largely assume simplified kinematics or full agent visibility. Toward realistic indoor deployment, Falcon [13] introduces the dynamic Social-HM3D benchmark via auxiliary motion prediction and obstruction penalties. Falcon summarizes observation history through a policy LSTM. SocialSTG complements this recurrent architecture with explicit pedestrian records that persist across visibility changes.

Spatio-Temporal Graph Networks and Interaction Memory. Spatio-temporal graphs (STGs) and generative models underpin multi-agent forecasting—including STGAT [17], Social-STGCNN [30], Trajectron++ [18, 33], LaneGCN [23], and diffusion predictors (MID [14], LED [28]). Although recent works handle missing data via trajectory imputation (MS-TIP [9]), streaming context alignment (RealMotion [39]), or spatial scene memory (SnapMem [49]), these models primarily support open-loop prediction and static exploration. Closed-loop navigation additionally requires handling abrupt FoV dropouts, decaying record reliability, and changing node-slot assignments. SocialSTG bridges this gap by embedding an uncertainty-gated persistent STG memory directly into a closed-loop visual policy.

3. Method

We formulate visual social PointGoal navigation as reaching target coordinates among dynamic pedestrians from egocentric RGB-D images, relative goal coordinates, and the previous action. As shown in Fig. 2, SocialSTG ex-

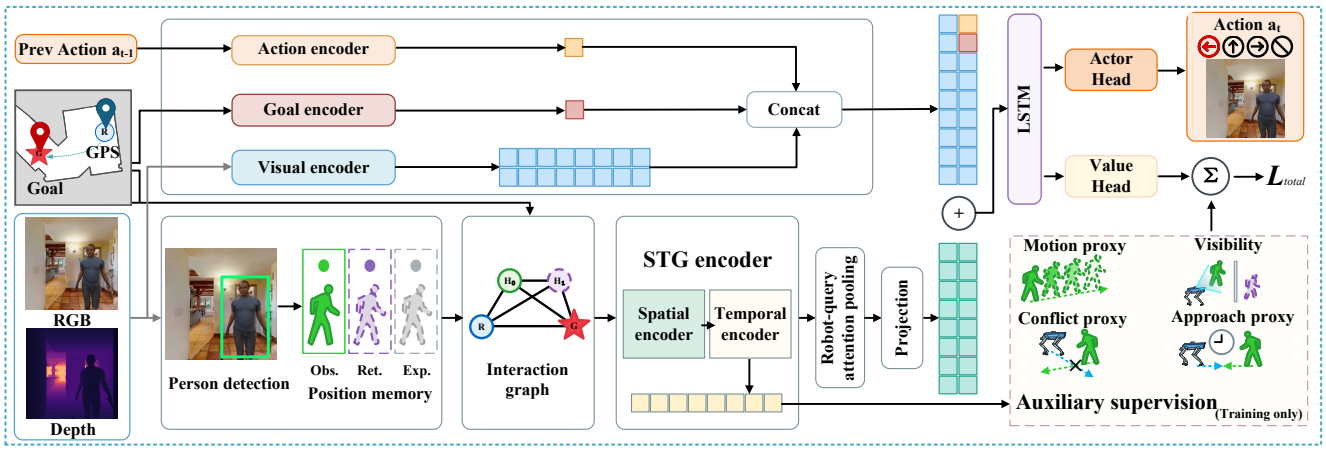


Figure 2. **Overview of SocialSTG.** Shared RGB-D observations feed parallel visual and graph streams. A position memory bank tracks YOLOv8n detections and retained pedestrian records to build the interaction graph processed by the STG encoder (Fig. 4). In the position-memory and interaction-graph panels, green indicates observed records, purple indicates retained records, and grey indicates expired records. Solid and dashed node outlines distinguish observed and retained states, respectively. Projected graph features are added to the concatenated visual, goal, and previous-action representation before the policy LSTM. Auxiliary heads provide training supervision for relative motion, observation status, conflict, and approach-time proxies.

tends Falcon [13] by pairing its visual backbone with an explicit spatio-temporal interaction graph. Whereas the visual stream encodes raw scene appearance and geometry, the graph branch tracks individual pedestrians—both visible and recently occluded—and models their evolving relations with the robot and goal. At each step, front-end detections update a persistent memory bank to construct the interaction graph (Sec. 3.1). An STG encoder then applies spatial attention and uncertainty-gated recurrence to update record-indexed node states (Sec. 3.2), which are pooled, projected, and added to the concatenated policy input before the policy LSTM (Sec. 3.3). We initialize the visual policy from the same pretrained PointNav weights used to initialize Falcon and jointly optimize the visual policy and graph network using the navigation reward alongside training-only auxiliary graph supervision (Sec. 3.4).

3.1. Dynamic Interaction Graph

Representing people across observations. At step t, the interaction graph $\mathcal{G}_t = (V_t, E_t)$ comprises the robot (R), goal (G), and a bounded set of human nodes (H_k, indexed by memory-bank record k), where visible detections and latent tracks share capacity. Front-end detections use YOLOv8n, and depth measurements provide two-dimensional position proxies. For a detection with depth estimate d and normalized horizontal box-center offset b, the stored pedestrian position is $\mathbf{p} = [d, bd]^\mathsf{T}$. The memory bank associates each incoming observation with the nearest stored position and assigns a dedicated record ID. Index i denotes a graph slot whose mapping to bank records evolves dynamically. We use $\mathcal{G}_t$ for the interaction graph and G for the goal node. Human nodes store position, velocity, heading, bounding-box dimensions, depth confidence, observation age, and reliability cues; the goal node stores relative distance and bearing with zero position fields.

Unmatched tracked pedestrians are retained as latent nodes for a bounded lifetime. Stored positions propagate via $\mathbf{p}_i^{t+1} = \mathbf{p}_i^t + \mathbf{v}_i^t$ (using unit update increments), while age and uncertainty increase and visibility decays. Subsequent matching observations reset age and latent uncertainty. This lifecycle keeps recently unobserved pedestrian records available to the policy. Crucially, the memory bank and recurrent states serve distinct roles: the bank governs record accessibility, whereas recurrence preserves learned interaction history, allowing unobserved tracks to remain active in graph reasoning.

Node insertion, retention, and removal. Figure 3 illustrates the lifecycle of pedestrian records using saved episode observations and graph states, including observed, retained, and expired records. Matched detections update the corresponding record, mark it as observed, and reset its age and latent uncertainty; unmatched detections initialize new bank IDs. Because association precedes age-based pruning, records at the retention limit can still be refreshed. Unmatched records remain active while latent memory is enabled and their incremented age stays within the retention threshold (20 updates in our setup). Capacity (up to 8 human records) is enforced by prioritizing observed over latent records, then younger records, breaking ties by bank ID. Excess records are pruned, and episode resets clear the bank.

These operations maintain the active record set and align

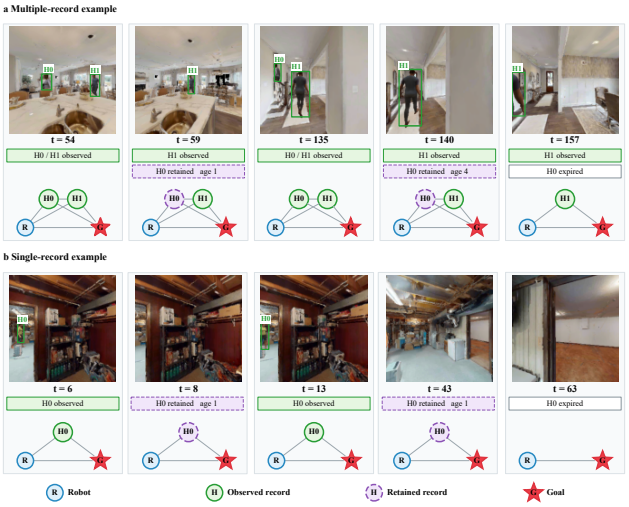


Figure 3. **Persistence and expiry of human records.** Each column pairs an RGB observation with record status and the corresponding interaction graph. (a) With multiple records, H_0 is retained when unmatched, updated when observed again, and eventually removed after expiry, while H_1 remains active in the displayed graphs. (b) A single record alternates between observed and retained states before expiring, leaving only the robot and goal nodes. Green solid and purple dashed outlines denote observed and retained records; age counts updates since the last matched observation. Blue circles and red stars denote the robot and goal. Frames are nonconsecutive, and H_k denotes a memory-bank record rather than a verified pedestrian identity.

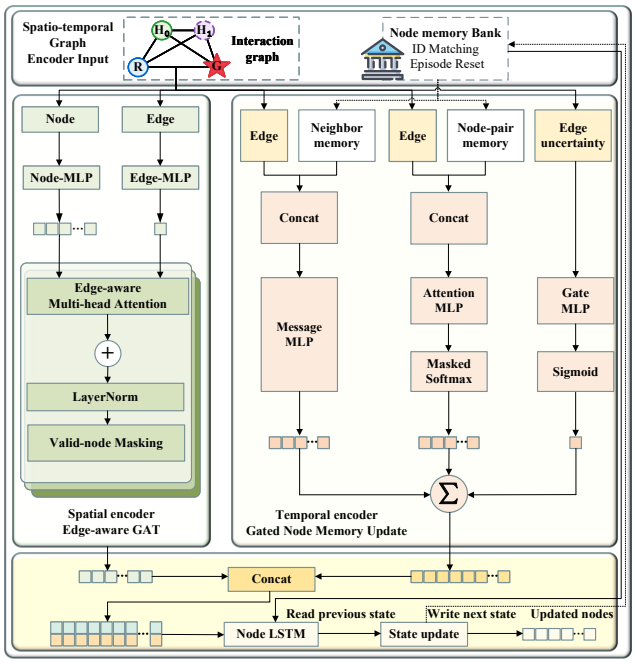


Figure 4. **Spatio-temporal graph encoding module.** Spatial graph attention aggregates interactions among nodes, while uncertainty-gated temporal message passing incorporates historical information. The node memory bank aligns hidden states by record ID, and a node-wise LSTM updates the aligned states.

each retained record with its graph state. Record insertion or expiration adds or removes human nodes and incident edges, whereas observed-to-latent transitions modify node attributes while preserving connectivity. Robot and goal nodes persist across updates (unused human slots are masked), and message gating scales contributions along the active edges.

Connectivity and reliability. Each pair of distinct active nodes exchanges messages in both directions, with $E_t = \{(i,j) : i,j \in V_t, i \neq j\}$. We index the message from sender j to receiver i by (i,j). The adjacency is symmetric, while ordered edge features and receiver-specific attention permit asymmetric message contributions. Track insertion, removal, and kinematic updates refresh the node set and edge attributes. Node reliability cues $\mathbf{u}_i = [u_i^{\text{depth}}, u_i^{\text{track}}, u_i^{\text{occ}}, u_i^{\text{latent}}]$ capture depth validity/variance, spatial association distance, and time since last observation (depth and initial occlusion uncertainty share a unified confidence signal). For uncertainty component $\ell \in \{\text{depth}, \text{track}, \text{occ}, \text{latent}\}$, each edge stores $u_{ij}^{\ell} = \max(u_i^{\ell}, u_j^{\ell})$. It also stores the mean across these four components, $\bar{u}_{ij} = \frac{1}{4}\sum_{\ell} u_{ij}^{\ell}$, conditioning message passing on observation quality and track age. The vector $\mathbf{u}_{ij}$ contains the four endpoint-maximum components; $\bar{u}_{ij}$ is an additional scalar edge feature.

Motion-related edge features. Edges encode proxy distance, bearing, velocity differences, visibility, reliability, and short-horizon kinematic features. Let $\mathbf{p}_i$ and $\mathbf{v}_i$ denote stored position and velocity fields, with offsets $\mathbf{r}_{ij} = \mathbf{p}_j - \mathbf{p}_i$ and $\mathbf{w}_{ij} = \mathbf{v}_j - \mathbf{v}_i$. Writing $s_{ij} = \|\mathbf{w}_{ij}\|_2$ and $[x]_0^{10} = \min(10, \max(0, x))$, we compute:

$$\tau_{ij} = \begin{cases} 10, & s_{ij} < 10^{-6}, \\ \left[-\dfrac{\mathbf{r}_{ij}^{\mathsf{T}} \mathbf{w}_{ij}}{\max(s_{ij}^2, 10^{-6})} \right]_0^{10}, & \text{otherwise}, \end{cases} \tag{1}$$

$$d_{ij}^{*} = \max(0, \|\mathbf{r}_{ij}\|_2 - \min(\tau_{ij}, 5) s_{ij}) \,. \tag{2}$$

The time proxy defaults to 10 for negligible velocity differences and is capped at 10 otherwise. The scalar d_{ij}^{*} is a distance-reduction heuristic based on the current separation and relative speed over a capped horizon. Two additional channels encode exponential proximity decay $\exp(-d_{ij}^{*})$ and a piecewise conflict score: 1 if $d_{ij}^{*} < 0.75$, and $\max(0, 1 - d_{ij}^{*}/2.5)$ otherwise. These heuristic channels depend on the stored feature representation and the quality of detection and depth estimation.

3.2. Spatio-temporal Graph Encoding

The STG encoder (Fig. 4) extracts spatial context before updating record-indexed recurrent states.

Spatial encoding. Node and edge MLPs project raw attributes into stacked multi-head graph-attention layers to produce spatial representations $\mathbf{z}_i^t$. Attention logits combine query–key compatibility with learned edge biases operating on encoded channels, while value messages fuse node and edge projections.

Gated temporal encoding. To preserve temporal continuity across shifting graph slots, previous recurrent states $(\bar{\mathbf{h}}_i^{t-1}, \bar{\mathbf{c}}_i^{t-1})$ are remapped by persistent bank IDs. Combining raw edge features $\mathbf{e}_{ij}^t$ with prior hidden states yields:

$$\mathbf{m}_{ij}^t = \phi_m([\bar{\mathbf{h}}_j^{t-1}, \mathbf{e}_{ij}^t]), \tag{3}$$

$$\alpha_{ij}^t = \mathrm{softmax}_{j\in\mathcal{N}(i)}\, \phi_a([\bar{\mathbf{h}}_i^{t-1}, \bar{\mathbf{h}}_j^{t-1}, \mathbf{e}_{ij}^t]), \tag{4}$$

$$\gamma_{ij}^t = \sigma(\phi_u(\mathbf{u}_{ij}^t)), \tag{5}$$

$$\widetilde{\mathbf{m}}_i^t = \sum_{j\in\mathcal{N}(i)} \alpha_{ij}^t \gamma_{ij}^t \mathbf{m}_{ij}^t, \tag{6}$$

where $\mathcal{N}(i) = \{j \in V_t : j \neq i\}$ denotes active neighbors and ϕ_m, ϕ_a, ϕ_u are learned MLPs; square brackets denote concatenation. Parameterizing attention and gating independently decouples their roles: softmax attention α_{ij}^t allocates relative importance across neighbors, whereas the sigmoid gate γ_{ij}^t attenuates individual message magnitudes based on uncertainty $\mathbf{u}_{ij}^t$ prior to aggregation. The products $\alpha_{ij}^t\gamma_{ij}^t$ directly weight the summed messages. Since $\mathbf{u}_{ij}^t = \mathbf{u}_{ji}^t$, the shared gate yields $\gamma_{ij}^t = \gamma_{ji}^t$; directional differences arise through the messages and attention weights.

An LSTM cell [16] then updates the node state:

$$(\mathbf{h}_i^t, \mathbf{c}_i^t) = \mathrm{LSTMCell}([\mathbf{z}_i^t, \widetilde{\mathbf{m}}_i^t], \bar{\mathbf{h}}_i^{t-1}, \bar{\mathbf{c}}_i^{t-1}). \tag{7}$$

Human hidden states track their bank IDs across visible and latent phases (newly instantiated IDs initialize to zero). Robot and goal states reset to zero at each graph update, leaving long-term navigation history to the separate policy LSTM. Episode termination resets all states.

3.3. Graph–Visual Fusion and Action Prediction

Robot-query attention pools updated human and goal embeddings using the updated robot state as the query. The pooled value is concatenated with the projected robot state and linearly mapped to the summary vector $\mathbf{g}_t$. In parallel, the visual stream encodes egocentric RGB-D, goal, and previous-action inputs into $\mathbf{f}_t^{\mathrm{base}}$ (Fig. 2). We inject the pooled graph summary residually before the policy LSTM:

$$\mathbf{f}_t = \mathbf{f}_t^{\mathrm{base}} + \psi(\mathbf{g}_t), \tag{8}$$

initializing the final linear layer of projection ψ to zero to preserve pretrained behavior at the start of training. The policy LSTM integrates $\mathbf{f}_t$ over time. Its output is passed to an action head that parameterizes the action distribution and a value head that estimates the expected discounted return.

3.4. Auxiliary Supervision of Human States

To supervise the information encoded in human-node representations $\mathbf{h}_i^t$, we attach four MLP heads before pooling. These heads supervise short-horizon kinematic proxies, observation status, conflict proxies, and approach-time proxies, backpropagating into shared human representations alongside PPO. During training, their losses jointly optimize the shared human-node representations with the navigation objective.

Motion-proxy supervision. The trajectory head predicts short-horizon targets derived from the stored position and velocity fields. Let $\Delta\mathbf{v}_i = \mathbf{v}_i - \mathbf{v}_r$, where $\mathbf{v}_r$ is the stored robot velocity field. For observed records with $\|\Delta\mathbf{v}_i\|_2 > 10^{-6}$, the targets are

$$\mathbf{y}_{i,k} = \mathbf{p}_i + k\Delta\mathbf{v}_i, \qquad k = 1, \ldots, K,$$

where K counts unit update steps. These targets provide constant-velocity extrapolation supervision in the stored feature representation. We minimize the mean Euclidean error over observed record–step pairs satisfying the relative-speed threshold.

Observation and interaction supervision. The observation-status head uses binary cross-entropy to distinguish records matched to current detections from retained records. For the interaction targets, let $s_i = \|\Delta\mathbf{v}_i\|_2$ and define the capped approach-time proxy

$$\tau_i = \begin{cases} 10, & s_i < 10^{-6}, \\ \left[-\dfrac{\mathbf{p}_i^\top \Delta\mathbf{v}_i}{s_i^2}\right]_0^{10}, & \text{otherwise.} \end{cases}$$

This auxiliary target uses the stored pedestrian position and velocity difference; it is distinct from the graph-edge proxy τ_{ij}. A second binary cross-entropy head predicts the conflict label, which is positive when $\|\mathbf{p}_i + \Delta\mathbf{v}_i \min(\tau_i, 5)\|_2 < 0.75$ and $\tau_i < 5$. The approach-time head regresses τ_i using mean squared error. Both interaction targets and the observation-status loss cover all active human records, including retained records and capped time targets, while masking padding. Together, these record-derived targets regularize human-node representations with motion, observation-status, conflict, and approach-time supervision.

We optimize the joint objective via PPO [34]:

$$\mathcal{L} = \mathcal{L}_{\mathrm{PPO}} + \sum_{q\in\mathcal{Q}} \lambda_q \mathcal{L}_q, \quad \mathcal{Q} = \{\mathrm{traj}, \mathrm{vis}, \mathrm{conf}, \mathrm{app}\}, \tag{9}$$

with balancing weights λ_q, architectural dimensions, and thresholds detailed in Sec. 4.1.

4. Experiments

4.1. Evaluation Setup

We evaluate on Social-HM3D and test zero-shot transfer to Social-MP3D following Falcon [13]. A reconstructed campus cafeteria further tests scene-specific adaptation.

Training and evaluation. Following the initialization protocol of Falcon [13], SocialSTG and all ablation variants use the same pretrained PointNav weights as Falcon. Each variant is trained on Social-HM3D with its designated component configuration. The full model is trained for $13,127,680$ steps using YOLOv8n detections and depth-based position estimation. We evaluate all methods across the same $1,087$ validation episodes on Social-HM3D and 317 zero-shot episodes on Social-MP3D under a strictly unified evaluation framework, using the authors' latest protocol evaluation for the Falcon Social-MP3D baseline.

Implementation details. The graph stores up to 8 human records and retains unmatched records for 20 updates. We use a three-layer, four-head graph encoder with 256-d node states and a two-layer, 512-d policy LSTM. The prediction horizon is $K = 5$, with auxiliary weights $(1.0, 0.2, 0.5, 0.3)$ for trajectory, visibility, conflict, and approach-time losses. Cafeteria adaptation uses PPO over $3,000$ training episodes, retaining each model's observation inputs and auxiliary objectives; learning rates are 2.5×10^{-4} for Falcon and 1.0×10^{-4} for SocialSTG.

Evaluation protocol and metrics. Success requires STOP within 0.2 m of the goal in the current episode. We correct an episode-state carryover in Falcon's evaluator that could allow proximity alone to satisfy success after an earlier episode had stopped. All main comparisons use reevaluated scores under the corrected criterion; continuous planners in Table 3 retain their native terminal-distance criterion. Following Falcon [13], we report SR, success-weighted path and time efficiency (SPL and STL), human collision rate (HCR), and personal-space compliance (PSC), measured using a 1 m separation threshold.

The STOP requirement matters particularly for Falcon. On Social-HM3D, its strict SR and reach-only diagnostic are 49.22% and 57.50%, compared with 56.03% and 56.49% for SocialSTG. On Social-MP3D, the corresponding pairs are $41.32\%/48.90\%$ and $53.00\%/53.31\%$. The smaller gap for SocialSTG indicates that reaching the goal region more often leads to valid termination. Reach-only success is reported here to distinguish arrival from stopping, rather than as a second table metric.

Table 1. **Navigation on Social-HM3D and zero-shot transfer to Social-MP3D.** Dataset groups follow the evaluation layout of Falcon [13]. SR requires an episode-local STOP within 0.2 m of the goal. All metrics are percentages.

Dataset	Method	SR↑	SPL↑	STL↑	PSC↑	HCR↓
Social-HM3D	A* [15]	45.26	44.41	45.25	90.15	54.28
	ORCA [42]	32.38	27.85	31.63	**92.92**	36.43
	Falcon [13]	49.22	44.03	49.13	89.55	41.86
	SocialSTG	**56.03**	**49.07**	**55.72**	88.66	**35.23**
Social-MP3D (zero-shot)	A* [15]	43.85	42.89	43.85	90.04	55.21
	ORCA [42]	38.17	33.79	37.36	**93.42**	44.16
	Falcon [13]	41.32	35.22	41.24	89.53	43.85
	SocialSTG	**53.00**	**45.81**	**52.99**	89.39	**35.33**

4.2. Comparison with Navigation Models

Performance on Social-HM3D. SocialSTG improves SR, SPL, and STL over Falcon while reducing human collisions (Table 1). The gain therefore extends beyond reaching the goal region: more encounters end in valid completion rather than contact. Compared with A*, SocialSTG combines better completion and success-weighted efficiency with fewer collision failures, supporting the benefit of modeling pedestrian interactions alongside goal progress. Relative to ORCA, it completes more tasks and achieves better efficiency with a slightly lower HCR. Its lower PSC, however, shows that these gains do not imply greater interpersonal clearance. The main advantage is completing navigation with fewer collision episodes, rather than maximizing every measure of social comfort.

Zero-shot transfer to Social-MP3D. Without target-domain adaptation, SocialSTG retains its advantage over Falcon, A*, and ORCA in completion, success-weighted efficiency, and HCR. Its completion declines less than Falcon's across datasets, while its HCR remains nearly unchanged. This suggests that the learned interaction representation remains useful beyond the training scenes, although the aggregate comparison does not isolate which component drives transfer. PSC stays close to Falcon and below both classical planners, preserving the distinction between collision avoidance and personal-space compliance.

4.3. Component Ablations

Table 2 shows how persistence, message filtering, and policy access contribute to the full model. Removing track memory produces the largest loss in completion and success-weighted efficiency and the highest collision rate among the ablations. An unmatched observation then immediately removes the human node, its incident relations, and its associated recur-

Table 2. **Component ablations on Social-HM3D.** All variants use the same PointNav initialization. *Motion Features*: four kinematic edge channels; *Uncertainty Gate*: temporal message gating; *Track Memory*: unmatched-record retention; *Graph*: graph features supplied to the policy (the graph branch and auxiliary losses remain). *Falcon Auxiliary Tasks*: replacement with Falcon auxiliary tasks. All metrics are percentages.

Variant	SR↑	SPL↑	STL↑	PSC↑	HCR↓
Full	**56.03**	**49.07**	**55.72**	88.66	35.23
w/o Motion Features	54.84	46.51	54.17	**89.31**	**35.10**
w/o Uncertainty Gate	52.58	45.73	52.36	88.44	36.21
w/o Track Memory	48.25	41.99	47.89	88.73	39.39
w/o Graph	53.36	44.35	52.82	88.36	37.75
Falcon Auxiliary Tasks	52.38	45.52	51.82	89.08	35.40

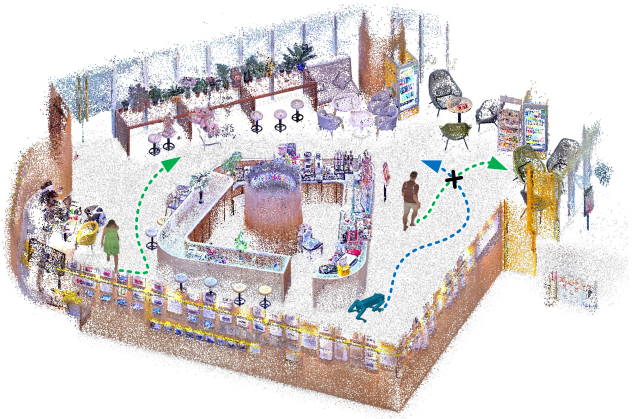


Figure 5. **Interaction context in the cafeteria digital twin.** Cutaway point-cloud rendering of the reconstructed scene with the quadruped robot and pedestrian models.

rent state, even when the pedestrian may still affect the next action. Retention bridges this interruption so that subsequent decisions can use recent interaction history and refresh it when a matching observation returns. Removing uncertainty gates also reduces completion and efficiency while increasing collisions. Spatial and temporal attention remain active in this variant: attention distributes relative weight among neighbors, whereas the separate gate can reduce the magnitude of a poorly supported message before it enters recurrent memory. The result is consistent with the need to regulate retained information, rather than treating persistence alone as sufficient. Removing motion-related edge channels lowers completion and efficiency but slightly improves HCR and PSC. These channels distinguish similar current separations with different relative motion, providing cues for coordinating progress through an encounter. Their observed benefit is thus primarily in task completion, not collision reduction; because SPL and STL are success-weighted, their decline does not by itself imply longer or slower successful paths. Suppressing the graph input to the policy reduces completion, efficiency, and PSC while increasing collisions, despite retaining graph computation and auxiliary losses. This separates learning graph representations from actually supplying their state-dependent summary during action selection: training the branch alone does not recover the benefit of graph–visual fusion. Finally, replacing the graph auxiliary objectives with Falcon's auxiliary tasks reduces completion and efficiency, with little change in collisions and a higher PSC. Supervising motion, observation status, conflict, and approach-time cues directly at human-node states appears better aligned with the representation later pooled by the policy. This comparison favors the proposed supervision over the original tasks, but does not establish the necessity of auxiliary learning itself or the contribution of each head. Overall, the ablations support complementary roles for keeping relevant records available, filtering their messages, and exposing them to the policy. These mechanisms explain the observed trends, while repeated-seed and targeted encounter analyses would be needed to establish their individual causal effects more precisely.

4.4. Cafeteria Digital Twin and Qualitative Analysis

The cafeteria digital twin tests navigation through shared walkways and counter-side passages (Fig. 5). Zero-shot Falcon and SocialSTG both struggle under the strict STOP criterion, indicating that the benchmark-trained policies do not directly resolve this scene's demands. The continuous planners in Table 3 use their native terminal-distance criterion, so their success and success-weighted efficiency scores provide complementary deployment context rather than a common-criterion ranking.

Scene-specific training improves completion and efficiency and reduces collisions for both Falcon and SocialSTG. Under the shared strict criterion, adapted SocialSTG retains an advantage in SR, SPL, STL, and HCR, suggesting that the benefit is not limited to the unadapted comparison. Falcon(seen) has higher PSC, however: the graph-augmented policy's advantage again concerns completion with fewer collision failures rather than the greatest personal-space margin. These results concern adaptation within the reconstructed cafeteria and do not establish transfer to new physical environments.

The synchronized cases in Fig. 6 show successful progress through pedestrian encounters and failures when separation becomes too small. Successful runs combine continued route progress with clearance through the shared passage; the failures expose limits in handling close encounters despite retained interaction context. The broader case set also contains stalled progress and premature stopping, showing that collision avoidance alone does not ensure completion. Appendix A gives episode IDs, frame indices, and detailed outcome evidence.

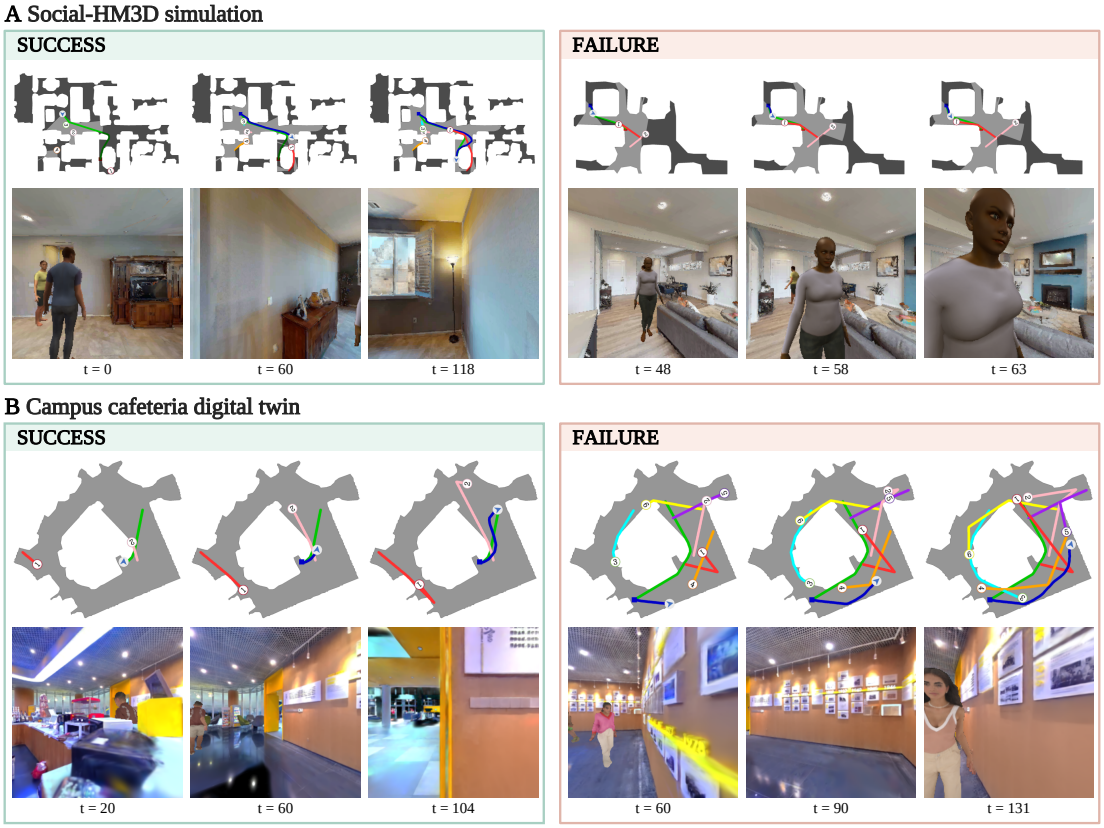


Figure 6. **Success and failure qualitative cases.** Synchronized egocentric RGB views and top-down maps across Social-HM3D (A) and the cafeteria digital twin (B), illustrating navigation outcomes and interaction dynamics.

Table 3. **Navigation in the cafeteria digital twin.** Comparison of classical planners, zero-shot transfer, and scene-specific adaptation (*seen*) across 100 minival episodes. †NavDP and LoGoPlanner are evaluated under their official success criterion (terminal Euclidean distance to goal < 1.0 m), whereas all other methods enforce the strict 0.2 m STOP condition.

Method	SR↑	SPL↑	STL↑	PSC↑	HCR↓
A* [15]	41.00	40.23	41.00	89.24	59.00
ORCA [42]	42.00	38.02	41.80	89.17	51.00
NavDP† [2]	43.00	38.40	43.00	90.67	41.00
LoGoPlanner† [32]	12.00	9.24	10.82	89.24	61.00
Falcon [13]	5.00	3.54	4.09	89.79	59.00
SocialSTG	10.00	7.57	9.17	87.48	57.00
Falcon(seen) [13]	59.00	49.32	59.00	**91.90**	33.00
SocialSTG (seen)	**65.00**	**55.49**	**65.00**	90.74	**30.00**

5. Conclusion

We presented SocialSTG, which maintains recent pedestrian information through persistent graph memory. Retention bridges observation gaps, uncertainty-gated recurrence regulates temporal updates, and graph fusion supplies interaction context to the policy. Across Social-HM3D and zero-shot Social-MP3D, it improves completion and success-weighted efficiency while reducing collision episodes relative to the evaluated baselines. Ablations support retention, gating, and graph input, while motion features and auxiliary supervision mainly benefit completion and efficiency. Scene-specific adaptation retains an advantage over adapted Falcon in the cafeteria digital twin. Personal-space compliance remains a tradeoff. Future work will improve pedestrian association and test whether these gains extend to physical environments.

References

[1] Amir Bar, Gaoyue Zhou, Danny Tran, Trevor Darrell, and Yann LeCun. Navigation world models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. 2

[2] Wenzhe Cai, Jiaqi Peng, Yuqiang Yang, Yujian Zhang, Meng Wei, Hanqing Wang, Yilun Chen, Tai Wang, and Jiangmiao Pang. NavDP: Learning sim-to-real navigation diffusion policy with privileged information guidance. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 5233–5240, 2026. 2, 8

[3] Devendra Singh Chaplot, Dhiraj Gandhi, Saurabh Gupta, Abhinav Gupta, and Ruslan Salakhutdinov. Learning to explore using active neural SLAM. In *International Conference on Learning Representations (ICLR)*, 2020. 2

[4] Changan Chen, Yuejiang Liu, Sven Kreiss, and Alexandre Alahi. Crowd-robot interaction: Crowd-aware robot navigation with attention-based deep reinforcement learning. In *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, pages 6015–6022, 2019. 1, 2

[5] Changan Chen, Sha Hu, Payam Nikdel, Greg Mori, and Manolis Savva. Relational graph learning for crowd navigation. In *Proc. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 10007–10013, 2020. 2

[6] Changan Chen et al. Socialnav: Training human-inspired foundation model for socially-aware embodied navigation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026. 2

[7] Yu Fan Chen, Michael Everett, Miao Liu, and Jonathan P. How. Socially aware motion planning with deep reinforcement learning. In *Proc. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 1343–1350, 2017. 2

[8] Cheng Chi, Zhenjia Xu, Siyuan Feng, Eric Cousineau, Yilun Du, Benjamin Burchfiel, Russ Tedrake, and Shuran Song. Diffusion policy: Visuomotor policy learning via action diffusion. *The International Journal of Robotics Research (IJRR)*, 2024. 2

[9] Pranav Singh Chib, Achintya Nath, Paritosh Kabra, Ishu Gupta, and Pravendra Singh. MS-TIP: Imputation aware pedestrian trajectory prediction. In *Proceedings of the International Conference on Machine Learning (ICML)*, pages 8562–8578, 2024. 2

[10] Daniel Flögel, Lars Fischer, Thomas Rudolf, Tobias Schürmann, and Sören Hohmann. Socially integrated navigation: A social acting robot with deep reinforcement learning. In *Proc. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 4785–4792, 2024. 2

[11] Anthony Francis, Claudia Pérez-d'Arpino, Chengshu Li, Fei Xia, Alexandre Alahi, Rachid Alami, Aniket Bera, Abhijat Biswas, Joydeep Biswas, Rohan Chandra, et al. Principles and guidelines for evaluating social robot navigation algorithms. *ACM Transactions on Human-Robot Interaction (THRI)*, 14 (2):1–65, 2025. 1

[12] Samiran Gode, Abhijeet Nayak, Débora N. P. Oliveira, Michael Krawez, Cordelia Schmid, and Wolfram Burgard. FlowNav: Combining flow matching and depth priors for efficient navigation. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025. 2

[13] Zeying Gong, Tianshuai Hu, Ronghe Qiu, and Junwei Liang. From cognition to precognition: A future-aware framework for social navigation. In *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, pages 9122–9129, 2025. 1, 2, 3, 6, 8

[14] Tianpei Gu, Guangyi Chen, Junlong Li, Chunze Lin, Yongming Rao, Jie Zhou, and Jiwen Lu. Stochastic trajectory prediction via motion indeterminacy diffusion. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 17113–17122, 2022. 2

[15] Peter Hart, Nils Nilsson, and Bertram Raphael. A formal basis for the heuristic determination of minimum cost paths. *IEEE Transactions on Systems Science and Cybernetics*, 4(2): 100–107, 1968. 6, 8

[16] Sepp Hochreiter and Jürgen Schmidhuber. Long short-term memory. *Neural Computation*, 9(8):1735–1780, 1997. 5

[17] Yingfan Huang, Huikun Bi, Zhaoxin Li, Tianlu Mao, and Zhaoqi Wang. STGAT: Modeling spatial-temporal interactions for human trajectory prediction. In *Proc. IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 6271–6280, 2019. 2

[18] Boris Ivanovic and Marco Pavone. The trajectron: Probabilistic multi-agent trajectory modeling with dynamic spatiotemporal graphs. In *Proc. IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 2375–2384, 2019. 2

[19] Haresh Karnan, Anirudh Nair, Xuesu Xiao, Garrett Warnell, Sören Pirk, Alexander Toshev, Justin Hart, Joydeep Biswas, and Peter Stone. Socially compliant navigation dataset (SCAND): A large-scale dataset of demonstrations for social navigation. *IEEE Robotics and Automation Letters (RA-L)*, 7(4):11807–11814, 2022. 2

[20] Kapil Katyal, Yuxiang Gao, Jared Markowitz, Sara Pohland, Corban Rivera, I-Jeng Wang, and Chien-Ming Huang. Learning a group-aware policy for robot navigation. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 11328–11335, 2022. 2

[21] Naman Kumar, Antareep Singha, Laksh Nanwani, Dhruv Potdar, Tarun R, Fatemeh Rastgar, Simon Idoko, Arun Kumar Singh, and K. Madhava Krishna. Crowdsurfer: Sampling optimization augmented with vector-quantized variational autoencoder for dense crowd navigation. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 16854–16860, 2025. 2

[22] Hongxin Li, uru Wang, Xuesong Yang, Zhaoxiang Wang, and Jun Tie. MemoNav: Working memory model for visual navigation. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 6681–6687, 2024. 2

[23] Ming Liang, Bin Yang, Rui Hu, Yun Chen, Renjie Liao, Song Feng, and Raquel Urtasun. Learning lane graph representations for motion forecasting. In *Proc. European Conference on Computer Vision (ECCV)*, pages 541–556, 2020. 2

[24] Shuijing Liu, Peixin Chang, Weihang Liang, Neeloy Chakraborty, and Katherine Driggs-Campbell. Decentralized structural-RNN for robot crowd navigation with deep

reinforcement learning. In *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, pages 3517–3524, 2021. 1, 2

[25] Shuijing Liu, Peixin Chang, Zhe Huang, Neeloy Chakraborty, Kaiwen Hong, Weihang Liang, D. Livingston McPherson, Junyi Geng, and Katherine Driggs-Campbell. Intention aware robot crowd navigation with attention-based interaction graph. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 12015–12021, 2023. 2

[26] Shangyi Luo, Ji Zhu, Peng Sun, Yuhong Deng, Cunjun Yu, Anxing Xiao, and Bo Xia. GSON: A group-based social navigation framework with large multimodal model. *IEEE Robotics and Automation Letters (RA-L)*, 10(10):10112–10119, 2025. 2

[27] Arjun Majumdar, Karmesh Yadav, Sergio Arnaud, Jason Ma, Claire Chen, Sneha Silwal, Aryan Jain, Vincent-Pierre Berges, Tingfan Wu, Jayesh Gupta, et al. Where are we in the search for an artificial visual cortex for embodied intelligence? In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 655–677, 2023. 2

[28] Weibo Mao, Chenxin Xu, Qi Zhu, Siheng Chen, and Yanfeng Wang. Leapfrog diffusion model for stochastic trajectory prediction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 5517–5526, 2023. 2

[29] Christoforos Mavrogiannis, Francesca Baldini, Allan Wang, Dapeng Zhao, Pete Trautman, Aaron Steinfeld, and Jean Oh. Core challenges of social robot navigation: A survey. *ACM Transactions on Human-Robot Interaction (THRI)*, 12(3):1–39, 2023. 1

[30] Abduallah Mohamed, Kun Qian, Mohamed Elhoseiny, and Christian Claudel. Social-STGCNN: A social spatio-temporal graph convolutional neural network for human trajectory prediction. In *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 14424–14432, 2020. 2

[31] Ye-Ji Mun, Masha Itkina, Shuijing Liu, and Katherine Driggs-Campbell. Occlusion-aware crowd navigation using people as sensors. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 12031–12037, 2023. 1, 2

[32] Jiaqi Peng, Wenzhe Cai, Yuqiang Yang, Tai Wang, Yuan Shen, and Jiangmiao Pang. Logoplanner: Localization grounded navigation policy with metric-aware visual geometry, 2025. 2, 8

[33] Tim Salzmann, Boris Ivanovic, Punarjay Chakravarty, and Marco Pavone. Trajectron++: Dynamically-feasible trajectory forecasting with heterogeneous data. In *Proc. European Conference on Computer Vision (ECCV)*, pages 683–700, 2020. 2

[34] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017. 5

[35] Dhruv Shah, Ajay Sridhar, Arjun Bhorkar, Noriaki Hirose, and Sergey Levine. GNM: A general navigation model to drive any robot. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 7226–7233, 2023. 2

[36] Dhruv Shah, Ajay Sridhar, Nitish Dashora, Kyle Stachowicz, Kevin Black, Noriaki Hirose, and Sergey Levine. ViNT: A foundation model for visual navigation. In *Conference on Robot Learning (CoRL)*, pages 711–733, 2023. 2

[37] Phani Teja Singamaneni, Pilar Bachiller-Burgos, Luis J. Manso, Anaís Garrell, Alberto Sanfeliu, Anne Spalanzani, and Rachid Alami. A survey on socially aware robot navigation: Taxonomy and future challenges. *The International Journal of Robotics Research (IJRR)*, 43(10):1533–1576, 2024. 1

[38] Daeun Song, Jing Liang, Amirreza Payandeh, Amir Hossain Raj, Xuesu Xiao, and Dinesh Manocha. VLM-Social-Nav: Socially aware robot navigation through scoring using vision-language models. *IEEE Robotics and Automation Letters (RA-L)*, 9(11):10456–10463, 2024. 2

[39] Nan Song, Bozhou Zhang, Xiatian Zhu, and Li Zhang. Motion forecasting in continuous driving. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. 2

[40] Ajay Sridhar, Dhruv Shah, Catherine Glossop, and Sergey Levine. NoMaD: Goal masked diffusion policies for navigation and exploration. In *IEEE International Conference on Robotics and Automation (ICRA)*, pages 63–70, 2024. 2

[41] Lei Tai, Jingwei Zhang, Ming Liu, and Wolfram Burgard. Socially compliant navigation through raw depth inputs with generative adversarial imitation learning. In *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, pages 1111–1117, 2018. 2

[42] Jur van den Berg, Stephen J. Guy, Ming Lin, and Dinesh Manocha. Reciprocal n-body collision avoidance. In *Robotics Research: The 14th International Symposium ISRR*, pages 3–19. Springer, 2011. 6, 8

[43] Erik Wijmans, Abhishek Kadian, Ari Morcos, Stefan Lee, Irfan Essa, Devi Parikh, Manolis Savva, and Dhruv Batra. DD-PPO: Learning near-perfect pointgoal navigators from 2.5 billion frames. In *International Conference on Learning Representations (ICLR)*, 2020. 2

[44] Erik Wijmans, Irfan Essa, and Dhruv Batra. VER: Scaling on-policy rl leads to the emergence of navigation in embodied rearrangement. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 7727–7740, 2022. 2

[45] Zhanteng Xie and Philip Dames. SCOPE: Stochastic cartographic occupancy prediction engine for uncertainty-aware dynamic navigation. *IEEE Transactions on Robotics (T-RO)*, 41:1420–1439, 2025. 1, 2

[46] Karmesh Yadav, Ram Ramrakhya, Arjun Majumdar, Vincent-Pierre Berges, Sachit Kuhar, Dhruv Batra, Alexei Baevski, and Oleksandr Maksymets. OVRL-V2: A simple state-of-art baseline for imagenav and objectnav. In *arXiv preprint arXiv:2303.07798*, 2023. 2

[47] Haodong Yang, Chenpeng Yao, Chengju Liu, and Qijun Chen. RMRL: Robot navigation in crowd environments with risk map-based deep reinforcement learning. *IEEE Robotics and Automation Letters (RA-L)*, 9(1):255–262, 2024. 1

[48] Ning Yang, Yan Huang, Kaiwen Peng, Ziheng He, Kai Wang, Cui Miao, Kailin Lyu, Guo Li, Xiaofeng Wang, Zheng Zhu,

et al. WAM-Nav: Asymmetric latent world-action modeling for unified visual navigation. In *Conference on Robot Learning (CoRL)*, 2026. 2

[49] Yuncong Yang, Han Yang, Jiachen Zhou, Peihao Chen, Hongxin Zhang, Yilun Du, and Chuang Gan. Snapmem: Snapshot-based 3d scene memory for embodied exploration and reasoning. *arXiv preprint arXiv:2411.17735*, 2024. 2

[50] Yiming Zeng et al. NaviDiffusor: Cost-guided diffusion model for visual navigation. *arXiv preprint arXiv:2504.10003*, 2025. 2

A. Detailed Case Evidence

Selection and replay. The original full-model selection contains 20 successes and 20 failures across 38 scenes. Its success strata comprise efficient completion, less efficient completion, and completion with contact. Its failure strata comprise contact-associated outcomes, time-limit outcomes, disagreements between $\mathrm{SR}_{\mathrm{reach}}$ and strict success, and an initially unclassified failure. These strata describe the observed outcomes. The subsequent export contains 40 complete rollouts and videos; seven outcome labels change, yielding 19 successes and 21 failures. Figure 6 instead presents four fresh multiview reruns, captured separately to obtain synchronized topdown and robot views. The diagnostic collection and illustrative reruns serve separate qualitative analyses. The cafeteria examples were recorded in the digital twin.

Synchronized multiview capture. Each frame contains robot RGB, depth, a rendered topdown view, a third-person view, a navigation-map view, physical poses, and outcome metrics. All views for a frame are rendered from the same simulator state with the action and pedestrian state held fixed. The grey-and-white topdown navigation map is placed directly above its matching RGB image. Separately saved terminal third-person views show the interaction region; they are omitted from the compact main-paper figure. The cafeteria terminal RGB mainly faces the wall, so contact is established jointly by the third-person view and the recorded collision metric. Indices start at the initial state; the last frame is the terminal state. Saved frame counts are 119 and 64 for the Social-HM3D success and failure, and 105 and 141 for the cafeteria success and failure.

Social-HM3D success: scene 7Ukhou1GxYi, episode 7. Frames 0, 60, and 118 show the new multiview run. The robot passes the initial encounter, continues toward the destination, and terminates successfully at a goal distance of approximately 0.076 m with zero human contact. The original diagnostic export used a different stochastic rollout and recorded track retention as visibility changed. The graph trace and multiview images provide complementary evidence from separate stochastic rollouts.

Social-HM3D failure: scene bxsVRursffK, episode 17. Frames 48, 58, and 63 show a close pedestrian encounter. The terminal state records a human collision while the goal remains distant. A seed-100 rerun succeeded; the displayed failure is the seed-101 rerun. Each displayed outcome is labeled using its saved rollout. The paired views document the spatial arrangement at contact.

Cafeteria success and failure: episodes 57 and 76. The new rerun of episode 72 ended in failure. We use the verified episode-57 success for synchronized visualization. Frames 20, 60, and 104 show progress around the counter and successful termination approximately 0.092 m from the goal with zero human contact. For episode 76, frames 60, 90, and 131 show navigation progress leading to a close pedestrian encounter before contact. The final displayed RGB/map pair is pre-terminal; the separately saved third-person view and collision metric are from the terminal state at frame 140. These indices identify saved simulator states. Physical poses and outcome metrics document the spatial progression and terminal outcome.

Additional failure modes. In scene XMHNu9rRQ1y, episode 0, the recorded position and goal distance remain unchanged from steps 100 through 499, while forward actions dominate. The robot remains stalled until the time limit, with zero human collisions. Static obstruction, control response, and local planning remain alternative explanations. In scene LEFTm3JecaC, episode 8, the replay ends with STOP at step 22 while the goal is still approximately 9.87 m away, identifying premature termination as the immediate failure mechanism. Cafeteria episode 27 also reaches the time limit with zero human collisions, adding a second example of stalled task completion.